

filter paper mass = 0,59 g

Data and calculations:

* Determination of the limiting reactant:

Data

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جامعة البلقاء التطبيقية

PART I

Choose the answer of the following question's and transfer it in CAPITAL letters to table

Q1: If You have left over some unused chemicals after you have completed the lab, you should:

- ☒ A- Place liquids and solids in the trash always.
- ☒ B- Follow all instructions given for disposal of chemicals.
- ☒ C- Put left over chemicals back into their original containers.
- ☒ D- Pour all the chemicals in the sink

Q2: According to solubility rules, which of the following statements is incorrect:

- ☒ A- Adding large amount of solute will show a precipitate because the solution reaches saturation.
- ☒ B- All Carbonate (CO_3^{2-}) salts are water insoluble.
- ☒ C- All Nitrate (NO_3^-) salts are water soluble.
- ☒ D- Increasing temperature of solvent will increase the solubility of solid solute.

Q3: A suitable equipment for measuring mass of liquid in density experiment is:

- ☒ A- Volumetric flask
- ☒ B- Pycnometer
- ☒ C- Balance
- ☒ D- Graduated cylinder

Q4: Measuring the boiling point of a compound can be used to _____

- ☒ A- Confirm the compound's formula
- ☒ B- Determine the compound's structure
- ☒ C- Calculate the compound's density
- ☒ D- Check the compound's purity

Q5: Suppose a boiling point experiment for a liquid is performed at the Dead sea, and on the top of Rum mountains. which of the following statements is correct regarding the measured boiling point of this liquid?

- ☒ A- The measuring boiling point will stay the same because it's the same compound.
- ☒ B- The boiling point will be lower at higher altitude due to the decreased atmospheric pressure.
- ☒ C- The boiling point will be higher at higher altitude due to the increased atmospheric pressure.
- ☒ D- The boiling point will be higher at higher altitude due to the decreased atmospheric pressure.

Q6: A sample of hydrate $\text{CoCl}_2 \cdot x\text{H}_2\text{O}$ is heated in a crucible. After heating, a blue powder of CoCl_2 is produced. If the mass percent of CoCl_2 in hydrate = 54.6%. Calculate the water of crystallization (x) of this salt? (Atomic weights: H=1, O=16, Cl=35.5, Co=59 g/mol)

☒ A- 6

B- 4

C- 5

D- 3

$$= \frac{45,4}{18} \frac{129}{54,6}$$

$$x = \frac{m}{m_r} \frac{m_r}{m}$$

Q7: Calculate the mass of water in $\text{CaSO}_4 \cdot 3\text{H}_2\text{O}$, if mass of anhydrous salt = 7.5g. (Molar mass of H_2O = 18, CaSO_4 = 136 g/mol).

A- 2.26g

B- 1.51g

C- 1.99g

☒ D- 2.98g

$$\frac{45,4}{18} \frac{129}{54,6}$$

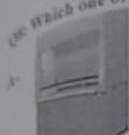
$$3 = \frac{m}{18} \frac{136}{7,5}$$

$$\frac{3(18)}{136} 7,5 = m$$

$$\frac{5856,6}{982,8}$$



Q1. Which one of these equipment used to do a quick medical treatment in lab?



first aid kit

PART 2:

Answer the following question's and transfer your answer to the table in the cover page

Q	Answer
1. This Pictogram stands for	Flammable
2. According to the following picture, the density of stone equals: Empty 90 g Liquid added 170 g Stone added 290 g <i>non polar</i>	$d = \frac{m}{V}$ $d = \frac{290 - 170}{140 - 100}$ $= \frac{120}{40}$ $= 3 \text{ g/ml}$
3. What physical observation do you see when I_2 is mixed with nonpolar solvent?	dissolving cubes
4. What is the way you use in your lab to heat flammable liquids?	put it in water bath then heat it
5. The name of this equipment, is	desiccator
6. How is the value of X (moles of water) of a hydrated salt affected if water is not completely driven off during heating? $\downarrow X = \frac{m}{M_r} \cdot \frac{M_r}{M}$	decrease
7. If a liquid sample contains some wood impurities, this cause the boiling point of liquid to: (increase, decrease, not affected)	not affect

Good LUCK